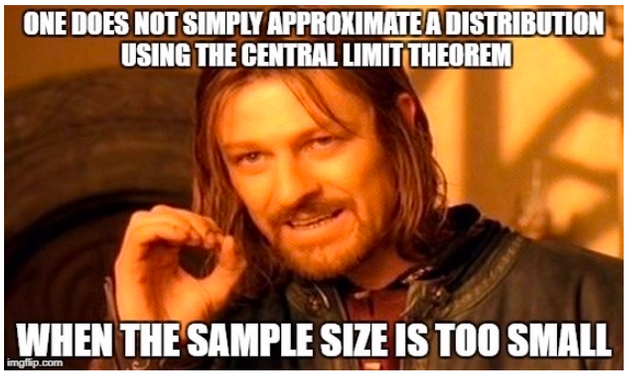
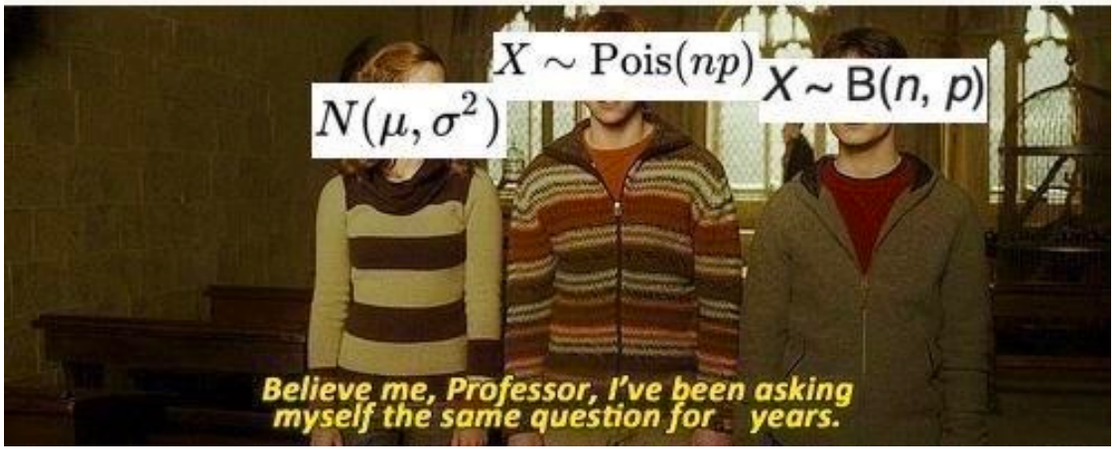
February 8, 2023.

Probability Theory Revision.

(Doubts, Revision, Interview Prep.)





Probability theory

1. **Random Experiment**: Outcome unknown.
2. **Sample Space**: Set of possible outcomes.
3. **Event**: Set; Subset of the sample space.
4. **Probability**: $\dfrac{\#\text{ outcomes in event set}}{\#\text{ outcomes in sample space}}$
5. **Conditional Probability**: Let A, B be two events.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

6. **Independence**: If $P(A|B) = P(A) \rightarrow$ A & B are independent.
7. **Bayes' theorem**: $P(A|B) = \dfrac{P(B|A) \cdot P(A)}{P(B)}$.
8. ——— $\rightarrow$ ${}^nP_r = \dfrac{n!}{(n-r)!}$

   $\rightarrow$ ${}^nC_r = \dfrac{{}^nP_r}{r!} = \dfrac{n!}{(n-r)! \cdot r!}$

The sample mean of the weights of the adult women of town A is larger than the sample mean of the weights of the adult women of town B. Moreover, the sample mean of the weights of the adult men of town A is larger than the sample mean of the weights of the adult men of town B. Can we conclude that the sample mean of the weights of the adults of town A is larger than the sample mean of the weights of the adults of town B? Explain your answer.

* The number of samples is not mentioned.

A					B		
Men	μ_{AM}	2	70	>	Men	μ_{BM}	10	69.
Women	μ_{AW}	10	60	>	Women	μ_{BW}.	2	59

$$\mu_A = \frac{70\times 2 + 60\times 10}{12} = \frac{140 + 600}{12} = \frac{740}{12}$$

$$\mu_B = \frac{69\times 10 + 59\times 2}{12} = \frac{690 + 118}{12} = \frac{808}{12}$$

$$\frac{740}{12} < \frac{808}{12}$$

A group of 5 boys and 10 girls is lined up in random order — that is, each of the 15! permutations are assumed to be equally likely.

a. What is the probability that the person in the 4th position is a boy?
b. What is the probability that the person in the 12th position is a boy?
c. What is the probability that a particular boy is in the 3rd position?

(a),(b) $\frac{5 \times 14!}{15 \; 15!}$

$\frac{5}{15} = \frac{1}{3}$.

(c) $1 \times \frac{14!}{15!} = \frac{1}{15}$

Of three cards, one is painted red on both sides; one is painted black on both sides; and one is painted red on one side and black on the other. A card is randomly chosen and placed on a table. If the side facing up is red, what is the probability that the other side is also red?

$$\{RR, BB, RB\}.$$

$$P(RR) = P(BB) = P(RB) = 1/3.$$

$$P(RR \mid \{RR, RB\}) = \frac{P(RR)\left(P(\{RR, RB\} \mid RR)\right)}{P(\{RR, RB\})}$$

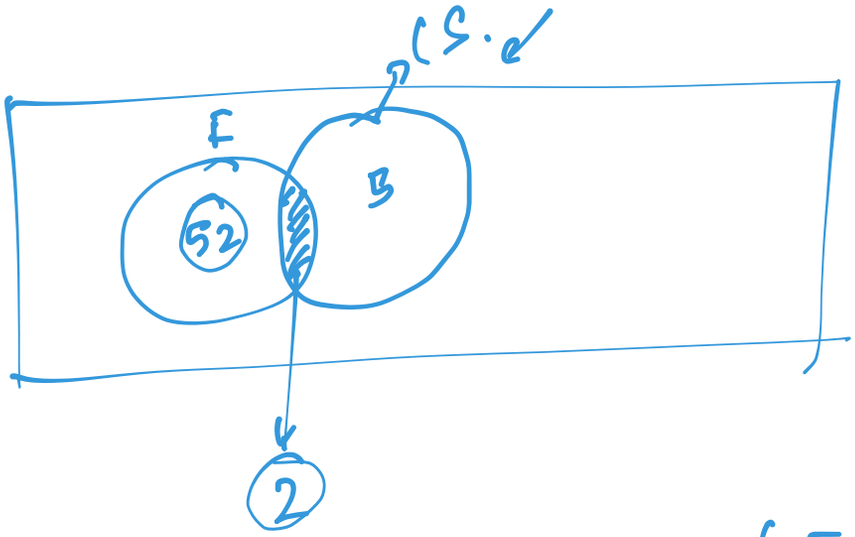
$$= \frac{1/3 \cdot 1}{1/2}$$

$$= \frac{2}{3}.$$

Fifty-two percent of the students at a certain college are females. Five percent of the students in this college are majoring in computer science. Two percent of the students are women majoring in computer science. If a student is selected at random, find the conditional probability that:

a. This student is female, given that the student is majoring in computer science.

b. This student is majoring in computer science, given that the student is female.



(a) $P(F \mid CS) = \dfrac{P(F \cap CS)}{P(CS)} = \dfrac{2}{5}$

(b) $P(CS \mid F) = \dfrac{P(F \cap CS)}{P(F)} = \dfrac{\cancel{2}}{\cancel{52}} = \dfrac{1}{26}$

There are two local factories that produce microwaves. Each microwave produced at factory A is defective with probability .05, whereas each one produced at factory B is defective with probability .01. Suppose you purchase two microwaves that were produced at the same factory, which is equally likely to have been either factory A or factory B. If the first microwave that you check is defective, what is the conditional probability that the other one is also defective?Out of 4 friends,

(a) Probability that all have the same birthday?

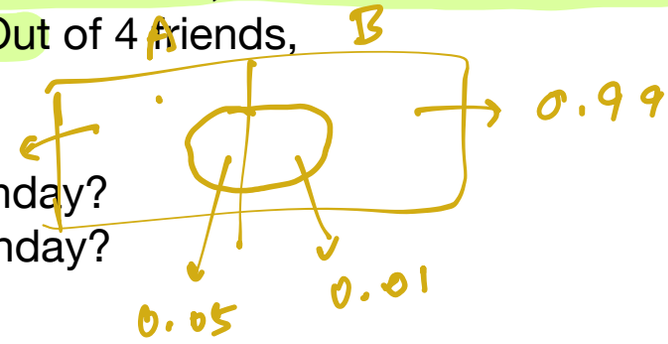
(b) Probability that all have a different birthday?

(c) Probability that 2 of them have the same birthday?

(d) Probability that 3 of them have the same birthday?

$P(D|A) = 0.05$

$P(D|B) = 0.01$



$P(A) = P(B) = 1/2.$

$D_1 \rightarrow$ Event that 1st microwave is defective.

$D_2 \rightarrow$ Event that 2nd microwave is defective.

$$P(D_2 | D_1) = \frac{P(D_1 \cap D_2)}{\boxed{P(D_1)}}$$

$$P(D_1) = P(D \cap A) + P(D \cap B)$$

$$= P(D|A) \cdot P(A) + P(D|B) \cdot P(B)$$

$$= 1/2\,(0.05 + 0.01) = 1/2 \times 0.06.$$

$P(D_1 D_2)$.

$D_A \quad D_A \qquad\qquad D_B \quad D_B$.

$$P(D_1 D_2) = P(D_A D_A) + P(D_B D_B)$$

$$P(D_1 D_2 \mid A) \cdot \underline{P(A)} + P(D_1 D_2 \mid B) \cdot \underline{P(B)}$$

$$(0.05)^2 \cdot 1/2 + (0.01)^2 \times 1/2$$

$$\frac{1}{2}(0.0025 + 0.0001) = \frac{1}{2}(0.0026)$$

$$\frac{\cancel{1/2}\ 0.0026}{\cancel{1/2}\ 0.06} = \frac{\cancel{26}}{\cancel{600}} = \underline{\underline{\frac{13}{300}}}.$$

Out of 4 friends, (Assume not a leap year).

(a) Probability that all have the same birthday?

(b) Probability that all have a different birthday?

(c) Probability that 2 of them have the same birthday?

(d) Probability that 3 of them have the same birthday?

Days : { 1, 2, 3, . . . , 365 }.

Denominator = $(365)^4$

(a) Numerator : 365, $P = 1/(365)^3$

(b) $^{365}C_4 \times 4! = \frac{365!}{361! \times 4!} \times 4!$

{37, 144}

Numerator = $365 \times 364 \times 363 \times 362$.

(c) (i) $^{365}C_2 \times \frac{4!}{2!} \times 3 = \frac{365!}{362! \times 3!} \times 4!$

$3 \times 2 \times (365 \times 364 \times 363)$

(ii) $^{365}C_2 \times \frac{4!}{2!2!} = (365 \times 364 \times 3)$

1d) _ _ _ _

$$^{365}C_2 \times 2 \times \frac{4!}{3!}$$

$$\frac{365 \times 364}{\cancel{2!}} \times \cancel{2} \times 4$$

$$(365 \times 364) \times 4.$$

(a) .365

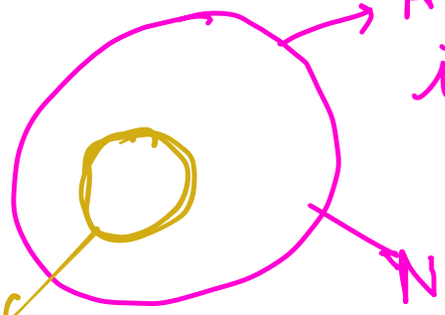
(b) .365 × 364 × 363 × 362.

(c) .(365 × 364 × 363) 6. + (365 × 364) 3

(d) . (365 × 364) × 4

$$= (365)^4$$

Walmart Case Study:



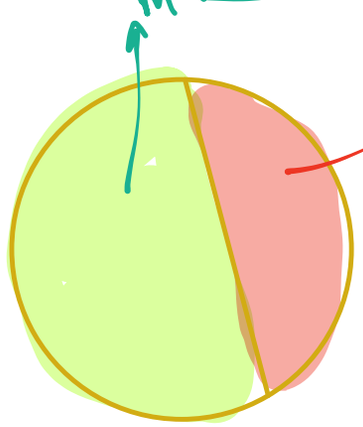
All possible purchases in Walmart on black friday.

Actually what we are observing.

Is the data we are observing:

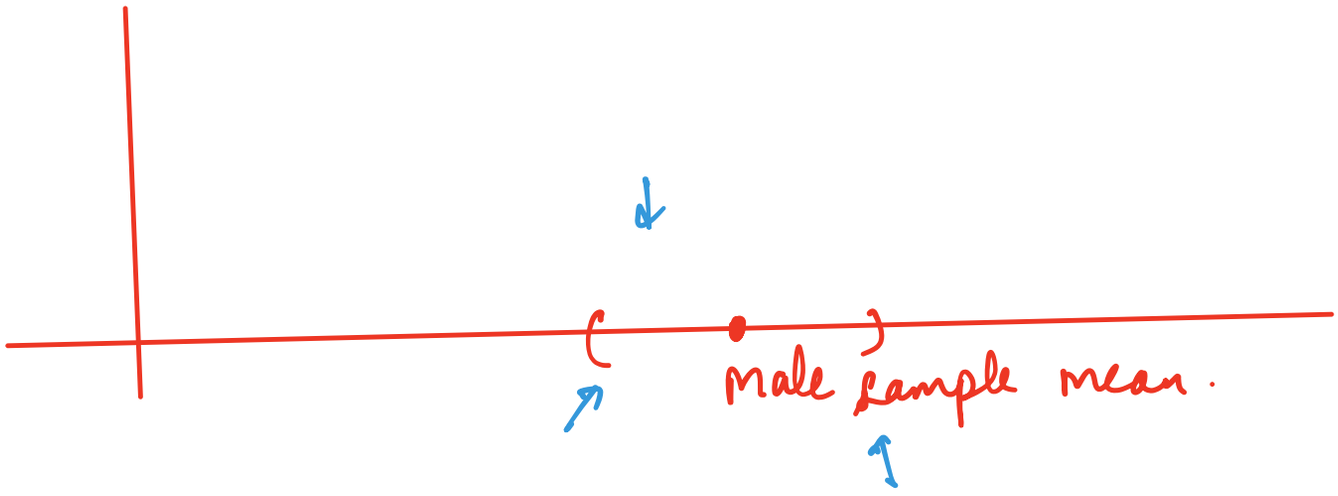
A. O

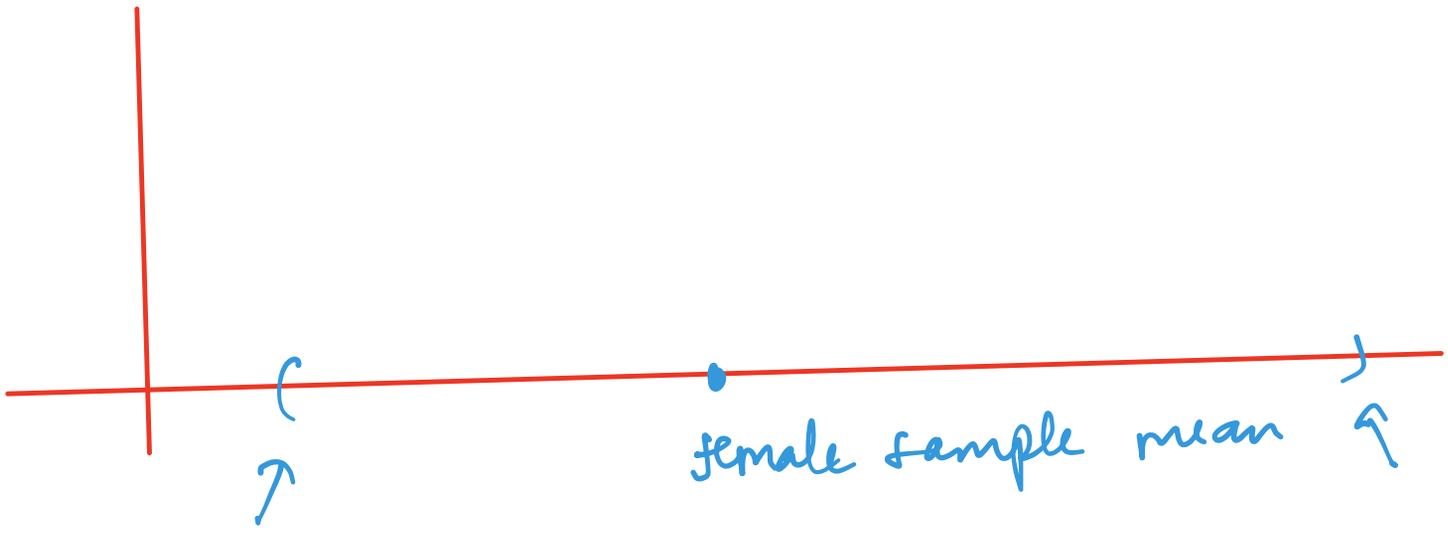
B. O



M → average of amount spent.

F. → average of amount spent.

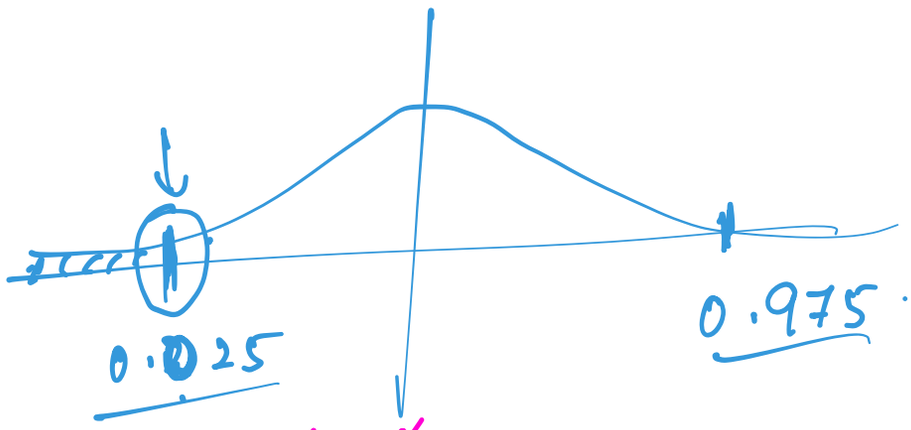

male sample mean.


female sample mean

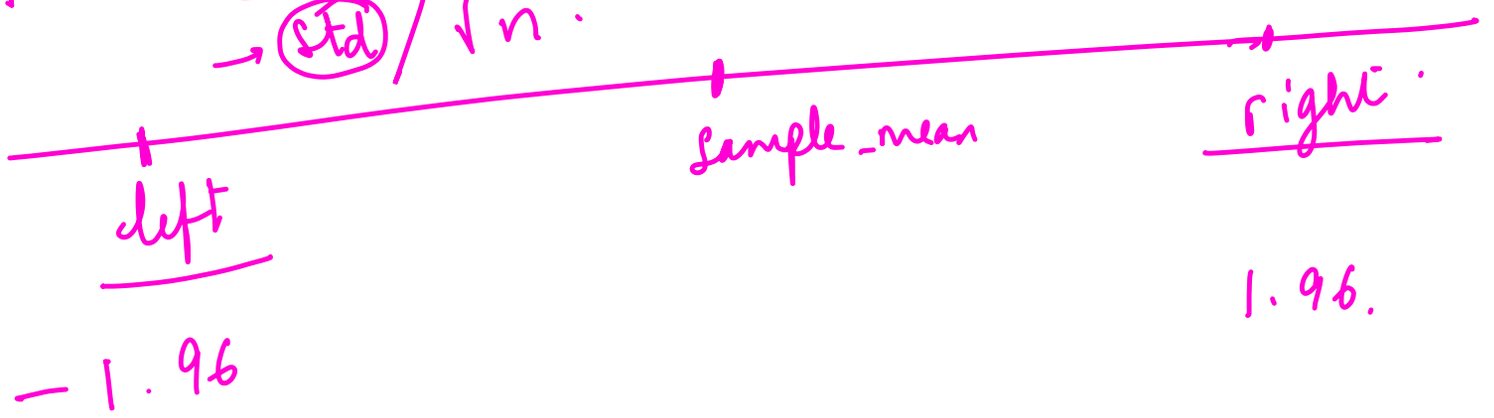
CLT. → 95%

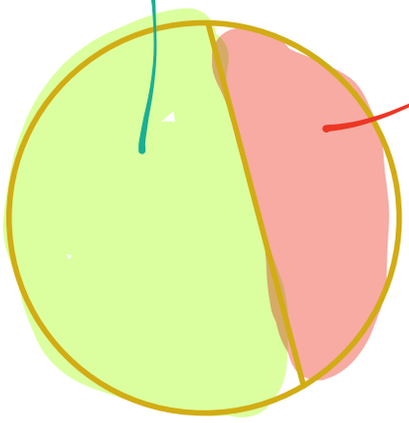
Z-value from ppf. = 1.96.

$$Z = \frac{\text{obs.} - \text{pop. mean}}{\text{pop. std} / \sqrt{n}}$$



$$-1.96 = \frac{\text{left} - \text{mean}}{\text{std} / \sqrt{n}}$$





$M \rightarrow F$

$$N \left\{ \begin{array}{l} [\ ,\ ,\ ,\ ,\ ] \rightarrow \text{size } n \rightarrow \bar{m}_1 \\ \vdots \\ [\ \ \ \ \ ] \rightarrow \bar{m}_N \end{array} \right.$$

sample with replacement.

$$N \left\{ \begin{array}{l} [\ ,\ ,\ \ \ ] \rightarrow \text{size } n \rightarrow \bar{m}_1 \\ \vdots \\ [\ \ \ \ \ ] \rightarrow \bar{m}_N \end{array} \right.$$

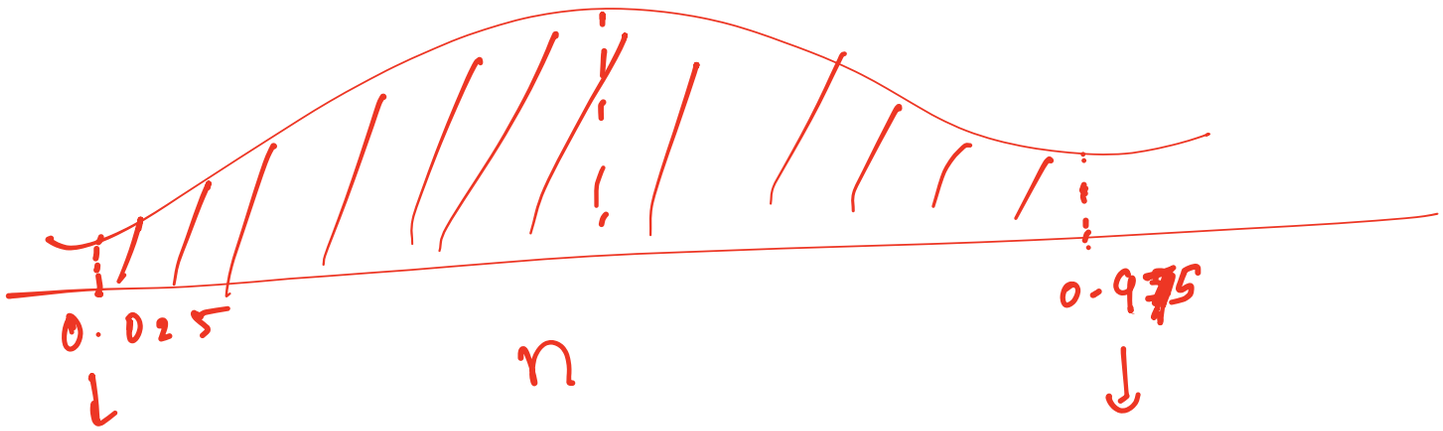
$n \rightarrow$ size of bootstraped sample.

$N \rightarrow$ number of repeats.

left 2.5^{th} percentile — right 97.5^{th} percentile.

limits.

In a certain experiment, for sample size n, the margin of error is 3%. How many more samples do we need to make the margin of error 0.3%? Note: Take the confidence interval to be 95%.



$$\frac{1.96}{10} \times \frac{\sigma}{\sqrt{n}}$$

$$1.96 \times \frac{\sigma}{\sqrt{100n}}$$

$100n \rightarrow 99n$

$$\frac{1.96}{10} \times \frac{\sigma}{\sqrt{n}}.$$

$$1.96 \times \frac{\sigma}{\sqrt{100n}}.$$